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ECA1004 - MPMC - Slot A

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12-10-25 CO2 - AT1 - Analytical Problem Solving

Q1. 2's Complement of a 16-Bit Number

Logic: Fetch the LSB and MSB from memory into a 16-bit register, invert all bits, add 1, and store it back in Little-Endian format.

Code:

```
MOV AX, [5000H]
NOT AX
ADD AX, 0001H
MOV [5200H], AX
HLT
```

eg. Number = 123HH

Original = 123H → 0001 0010 0011 0100 H
complement = EDCBH → 1110 1101 1100 1011 H
Add 1 = EDCCH → 1110 1101 1100 1100 H

Result LSB at 5200H = CCH ⇒ 1100 1100 H
Result MSB at 5201H = EDH ⇒ 1110 1101 H

Q2. Unsigned 8-bit Division

Logic: The 8-bit DIV instruction requires the 16-bit dividend in AX. We must set AH = 00H and AL = 90H (144 decimal). Dividing by BL = 1FH (31 decimal) yields a quotient of 4 and a remainder of 20.

Code :

```
MOV AL, 90H
MOV AH, 00H
MOV BL, 1FH
DIV BL
HLT
```

Calculation :

Convert to Decimal : 90H → 144

1FH → 31

Divide : 144 = 31 × 4 + 20

Register	Result	Meaning
AL	04H	Quotient = 4
AH	14H	Remainder = 20

Answer :

Quotient = 04H

Rembinder = 14H

Q3. 16 - Bit Unsigned Multiplication

Logic : Multiplying two 16-bit values generates a 32-bit product. The MUL instruction automatically multiplies AX by the operand (BX) and distributes the 32-bit result across the DX (higher-order) and AX (lower-order) registers.

Code :

```
MOV AX, 1111H
MOV BX, 0FFFFH
MUL BX
HLT
```

Calculation : $1111_H \times FFFF_H$

Using, $FFFF_H = 10000_H \div 1$

$\Rightarrow 1111_H \times (10000_H - 1)$

$\Rightarrow 11110000_H - 1111_H$

$\Rightarrow 1110EEEF_H$

Distribution of Result :

Register	Value	Properties
DX	1110H	Higher 16-bits
AX	EEEFH	Lower 16-bits

$\therefore DX : AX = 1110 : EEEF$

Q4. Block Transfer Using String operations

Logic : To efficiently copy a block of data, set the source and destination index registers, define the byte count in the loop counter (cx), clear the direction flag to auto-increment, and use a repeating string instruction [cite : 1, 2].

Code :

```
MOV AX, DS
MOV ES, AX
MOV SI, 7000H
MOV DI, 9000H
MOV CX, 0064H
CLD
L1 : MOVSB
LOOP L1
HLT
```

Calculation :

Convert Decimal 100 to Hexadecimal

Division	Quotient	Remainder
$100 \div 16$	6	4
$6 \div 16$	0	6

Read remained from bottom to top

$100_{10} = 64H$

Therefore,

$CX = 0064H$

Q5. Addressing Mode Analysis

No.	Instruction	Addressing Mode
1	ADD AX, 0100H	Immediate
2	SUB AX, [5000H]	Direct Memory
3	CMP BX, CX	Register
4	MOV AX, [BX] [SI]	Based - Index
5	MOV AX, 50H [BX][SI]	Based index with displacement

Calculation :

1. 2000H + 0100H ⇒ 2100H
2. 3000H - 1000H ⇒ 2000H
3. 2500H - 1500H ⇒ 1000H, flag updated
4. 2000H + 1000H ⇒ 3000H
5. 2000H + 1000H + 0050H ⇒ 3050H